

一、BDCBAD 二、1. 0 2. e^2 3. $2x \cos x^2 f(\sin x^2)$ 4. $-\frac{3}{2}$ 5. $y = x$ 6. $\frac{1}{2}x^2 + \frac{1}{4}x^4 + C$

$$\text{三、} \lim_{x \rightarrow 0} \frac{\sin x (e^x - \sqrt{1+2x})}{\ln(1+x)(1-\cos x)} = \lim_{x \rightarrow 0} \frac{e^x - (1+2x)^{\frac{1}{2}}}{x} = 2$$

$$\text{四、} \int_0^1 \frac{x^2}{\sqrt{2x-x^2}} dx = \int_0^1 \frac{x^2}{\sqrt{1-(x-1)^2}} dx = \int_{-\frac{\pi}{2}}^0 \frac{(1+\sin t)^2}{\sqrt{1-(\sin t)^2}} \cos t dt = \frac{3\pi}{4} - 2$$

五、设曲线方程为 $\begin{cases} x = 2(\ln \tan \frac{t}{2} + \cos t) \\ y = 2 \sin t \end{cases}$, 求 (1) $\frac{dy}{dx}, \frac{d^2y}{dx^2}$; (2) 曲线在 $t = \frac{\pi}{4}$ 对应点处的曲率.

$$(1) \frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \tan t, \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\frac{dy}{dx})}{x'(t)} = \frac{\sin t}{2 \cos^4 t} \quad (2) K = \frac{|y''|}{(1+y'^2)^{\frac{3}{2}}} = \frac{\frac{\sin t}{2 \cos^4 t}}{(1+\tan^2 t)^{\frac{3}{2}}} = \frac{1}{2}$$

六、如图所示, 曲线 $y = x^2$ 在 $[0, 1]$ 的一段与平行于 x 轴的直线 AB 交于 C , C 的横坐标为 η ($\eta \in [0, 1]$).

问 η 为何值时, 阴影部分面积 S 最小? 并求最小值.

$$S'(\eta) = \left(\eta^2 \int_0^\eta dx - \int_0^\eta x^2 dx + \int_\eta^1 x^2 dx - \eta^2 \int_\eta^1 dx \right)' = \left(\frac{4}{3} \eta^3 - \eta^2 + \frac{1}{3} \right)' = 4\eta^2 - 2\eta = 0, \eta = \frac{1}{2}$$

$$S_{\min}(\frac{1}{2}) = \frac{1}{4}, \quad S(0) = \int_0^1 x^2 dx = \frac{1}{3}, \quad S(1) = \int_0^1 (1-x^2) dx = \frac{2}{3}.$$

七、
$$\int \frac{\arctan e^x}{e^{2x}} dx = -\frac{1}{2} \int \arctan e^x d(e^{-2x}) = -\frac{1}{2} [e^{-2x} \arctan e^x - \int \frac{1}{e^{2x}(1+e^{2x})} de^x]$$

$$= -\frac{1}{2} (e^{-2x} \arctan e^x + e^{-x} + \arctan e^x) + C$$

八、设函数 $f(x)$ 在 $[0, 1]$ 上连续, 在 $(0, 1)$ 内可导, $f(0) = f(1) = 0, f(\frac{1}{2}) = 1$. 证明: 在 $(0, 1)$ 内至少存在一点 ξ , 使得 $f'(\xi) = 1$.

【证明】令 $F(x) = f(x) - x$. 则 $F(x)$ 在 $[0, 1]$ 上连续, 在 $(0, 1)$ 内可导. 又 $F(\frac{1}{2}) = \frac{1}{2}, F(1) = -1$, 由零点定理,

存在 $\xi_1 \in (\frac{1}{2}, 1)$, 使 $F(\xi_1) = 0$. $F(x)$ 在 $[0, \xi_1]$ 上满足罗尔定理条件, 则存在 $\xi \in (0, \xi_1)$, 使得 $F'(\xi) = 0$, 即 $f'(\xi) = 1$.

九、设数列 $\{x_n\}$ 满足: $0 < x_n < 1, (1-x_n)x_{n+1} > \frac{1}{4}, n \in N^+$. 证明: 数列 $\{x_n\}$ 收敛, 并求其极限.

【证明】

$$\frac{x_{n+1}}{x_n} > \frac{1}{4x_n(1-x_n)} > \frac{1}{4[x_n + (1-x_n)]^2} = 1, \quad \text{即 } x_{n+1} > x_n, \quad 0 < x_n < 1, \{x_n\} \text{ 单调增加有上界, 故收敛.}$$

$$\text{设 } \lim_{n \rightarrow \infty} x_n = a, a(1-a) = \frac{1}{4}, \lim_{n \rightarrow \infty} x_n = \frac{1}{2}$$

一、DBACCD 二、1. $\frac{1}{2}$ 2. $y-f(3)=3(x-3)$ 3. $\frac{n(n+1)}{2}n!$ 4. 2 5. $\frac{1}{2}$ 6. $\frac{\pi}{2}$

$$\text{三、} \lim_{x \rightarrow 0^-} \left(\frac{4+e^{\frac{1}{x}}}{1+e^{\frac{2}{x}}} + \frac{\tan 2x}{|x|} \right) = \frac{4+0}{1+0} - 2 = 2, \lim_{x \rightarrow 0^+} \left(\frac{4+e^{\frac{1}{x}}}{1+e^{\frac{2}{x}}} + \frac{\tan 2x}{|x|} \right) = \lim_{x \rightarrow 0^+} \left(\frac{4e^{\frac{2}{x}} + e^{-\frac{1}{x}}}{e^{\frac{2}{x}} + 1} + \frac{\tan 2x}{x} \right) = 2$$

$$\text{四、} \lim_{x \rightarrow 0} \frac{\int_{\cos x}^1 e^{-t^2} dt}{e^{x^2} - 1 + \tan^3 x} = \lim_{x \rightarrow 0} \frac{\int_{\cos x}^1 e^{-t^2} dt}{x^2} = \lim_{x \rightarrow 0} \frac{e^{-\cos^2 x} \sin x}{2x} = \frac{1}{2e}.$$

$$\text{五、} \text{设函数 } y = y(x) \text{ 由参数方程 } \begin{cases} x = 1 + 2t^2, \\ y = \int_1^{1+2\ln t} \frac{e^u}{u} du \end{cases} (t > 1) \text{ 所确定, 求 } \frac{d^2 y}{dx^2} \Big|_{t=e}.$$

$$\text{解: } \frac{dy}{dt} = \frac{e^{1+2\ln t}}{1+2\ln t} \cdot \frac{2}{t} = \frac{2et}{1+2\ln t}, \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e}{2(1+2\ln t)}, \frac{d^2 y}{dx^2} = -\frac{\frac{2}{t^2}e}{2(1+2\ln t)^2} \cdot \frac{1}{4t} = -\frac{e}{4t^2(1+2\ln t)^2}, \frac{d^2 y}{dx^2} \Big|_{t=e} = -\frac{1}{36e}.$$

六、设 $y = y(x)$ 是由方程 $e^{xy} - x + y - 2 = 0$ 确定的隐函数, 求 $y''(0)$.

解: 当 $x = 0$ 时, $y = 1$, $e^{xy}(y + xy') - 1 + y' = 0$, 解得 $y'(0) = 0$

$$e^{xy}(y + xy')^2 + e^{xy}(2y' + xy'') + y'' = 0 \text{ 解得 } y''(0) = -1.$$

七、 $\int \arctan \sqrt{x} dx = \int \arctan t \cdot dt^2 = (x+1) \arctan \sqrt{x} - \sqrt{x} + C$.

八、平面上通过点 $P(4, 9)$ 引一条直线, 使它在两个坐标轴上的截距都为正且其和最小, 求该直线方程.

$$\frac{x}{a} + \frac{y}{b} = 1, \frac{4}{a} + \frac{9}{b} = 1, b = 9 + \frac{36}{a-4}, \text{ 令 } l = a + b = a + 9 + \frac{36}{(a-4)^2}, \text{ 令 } l' = 0 \text{ 解得 } a = 10,$$

$$b = 15 \text{ 所以所求直线方程 } \frac{x}{10} + \frac{y}{15} = 1.$$

九、设函数 $f(x)$ 在 $[0, 1]$ 上有二阶导数, 且 $f(0) = f(1)$, $|f''(x)| \leq 1$. (1) 写出函数 $f(x)$ 在 x_0

$(x_0 \in (0, 1))$ 点处的具有 Lagrange 型余项的一阶 Taylor 公式; (2) 证明: 当 $x \in (0, 1)$ 时, $|f'(x)| \leq \frac{1}{2}$.

解: (1) $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(\xi)(x - x_0)^2$, 其中 ξ 在 x 与 x_0 之间.

$$(2) \begin{cases} f(0) = f(x_0) + f'(x_0)(0 - x_0) + \frac{1}{2}f''(\xi_1)(0 - x_0)^2, 0 < \xi_1 < x_0 \\ f(1) = f(x_0) + f'(x_0)(1 - x_0) + \frac{1}{2}f''(\xi_2)(1 - x_0)^2, x_0 < \xi_2 < 1 \end{cases} \Rightarrow f'(x_0) = \frac{1}{2}[f''(\xi_1)x_0^2 - f''(\xi_2)(1 - x_0)^2]$$

$$\text{两式相减, 得 } |f'(x_0)| \leq \frac{1}{2}[|f''(\xi_1)|x_0^2 + |f''(\xi_2)|(1 - x_0)^2] \leq \frac{1}{2}[x_0^2 + (1 - x_0)^2] \leq \frac{1}{2}[x_0^2 + (1 - x_0)^2] = \frac{1}{2}$$

一、CBDBAA 二、7. e 8. $-\pi dx$ 9. $y = \frac{1}{2}x + 1$. 10. $-x - \frac{x^2}{2} + \cdots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$.

11. $y = x - 3$ 12. 3

三、13. $\lim_{x \rightarrow 0} \frac{x - \arcsin x}{\sin^3 x} = -\frac{1}{6}$ 14. $\begin{cases} x = e^{-t} \\ y = \int_0^t \ln(1+u^2) du \end{cases}$, 求 $\left. \frac{d^2 y}{dx^2} \right|_{t=0}$. 解: 当 $t=0$ 时, $x=1, y=0$.

$$\frac{dy}{dx} = \frac{y'}{x'} = -e^t \ln(1+t^2). \quad \frac{d^2 y}{dx^2} = \frac{[-e^t \ln(1+t^2)]'}{-e^{-t}} = \frac{-e^t \ln(1+t^2) - \frac{2te^t}{1+t^2}}{-e^{-t}}, \quad \left. \frac{d^2 y}{dx^2} \right|_{t=0} = 0$$

$$15. \int \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1} dx = \int \frac{t-1}{t+1} 2t dt = t^2 - 4t + 4 \ln(1+t) + C_1 = x - 4\sqrt{x+1} + 4 \ln(1+\sqrt{x+1}) + C$$

$$16. \int_0^1 x \arcsin x dx = \left[\arcsin x \cdot \frac{x^2}{2} \right]_0^1 - \frac{1}{2} \int_0^1 x^2 \frac{1}{\sqrt{1-x^2}} dx = \frac{\pi}{4} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin^2 t}{\cos t} \cos t dt = \frac{\pi}{4} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^2 t dt = \frac{\pi}{8}$$

$$17. f(x) = \begin{cases} \frac{\varphi(x) - \cos x}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}, \varphi''(x) \text{ 连续且 } \varphi(0)=1. (1) \text{ 求 } a, \text{ 使得 } f(x) \text{ 连续}; (2) \text{ 求 } f'(0).$$

解: $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\varphi(x) - \cos x}{x} = \lim_{x \rightarrow 0} \frac{\varphi'(x) + \sin x}{1} = \varphi'(0)$, $a = \varphi'(0)$ 时 $f(x)$ 连续.

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{\varphi(x) - \cos x - x\varphi'(0)}{x^2} = \lim_{x \rightarrow 0} \left[\frac{1}{2} \frac{\varphi'(x) - \varphi'(0)}{x} + \frac{\sin x}{2x} \right] = \frac{1}{2} \varphi''(0) + \frac{1}{2}$$

18. 求函数 $f(x) = \int_0^{x^2} (2-t)e^{-t} dt$ 的最大值和最小值.

解: $f'(x) = 2x(2-x^2)e^{-x^2} = 0$, $x = 0, \pm\sqrt{2}$. $f_{\min}(0) = 0, f_{\max}(\pm\sqrt{2}) = \int_0^2 (2-t)e^{-t} dt = 1 + e^{-2}$

$$f(\pm\infty) = \int_0^{+\infty} (2-t)e^{-t} dt = \int_0^{+\infty} (t-2)de^{-t} = [(t-2)e^{-t}]_0^{+\infty} - \int_0^{+\infty} e^{-t} dt = 2 + [e^{-t}]_0^{+\infty} = 2 - 1 = 1$$

19. 设函数 $f(x)$ 在 $[0, 2]$ 上连续, 在 $(0, 2)$ 内可导, 且 $f(0) = f\left(\frac{1}{2}\right)$, $2 \int_{\frac{1}{2}}^1 f(x) dx = f(2)$, 证明:

存在一点 $\xi \in (0, 2)$, 使 $f''(\xi) = 0$.

证明: $\exists \eta \in \left[\frac{1}{2}, 1\right]$, 使 $f(2) = 2 \int_{\frac{1}{2}}^1 f(x) dx = f(\eta)$, $f(x)$ 在 $[\eta, 2]$ 罗尔即 $\xi_1 \in (\eta, 2)$, 使 $f'(\xi_1) = 0$.

$f(x)$ 在 $\left[0, \frac{1}{2}\right]$ 上罗尔即 $\xi_2 \in \left(0, \frac{1}{2}\right)$, 使 $f'(\xi_2) = 0$.

$f(x)$ 在 $[\xi_2, \xi_1]$ 上罗尔, 即 $\xi \in (\xi_2, \xi_1) \subset (0, 2)$, 使 $f''(\xi) = 0$.